

Capstone Project – Step 3: Plan of Frontend Specifications

1. Introduction

This document presents the frontend architecture, control flow, routing structure, and UI component plan for the BindFlow application. The frontend is designed to be clean, intuitive, and role-specific, providing an excellent user experience for both Clients and Staff/Admins. Proper planning at this stage ensures logical navigation, maintainable code, and efficient development.

2. Frontend Technology Stack

- Framework: React.js (Vite)
 - Routing: React Router v6
 - HTTP Client: Axios
 - Styling: Tailwind CSS
 - State Management: React Context API + useReducer
 - Form Handling: React Hook Form + Zod
 - Additional Libraries: jsPDF (stretch), react-signature-canvas (stretch)
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3. User Roles and Authentication Flow

- Public Routes (No authentication required): Landing Page, Login, Register
- Client Role: Can submit orders and track their own jobs
- Staff/Admin Role: Full access to manage orders, quotes, and production

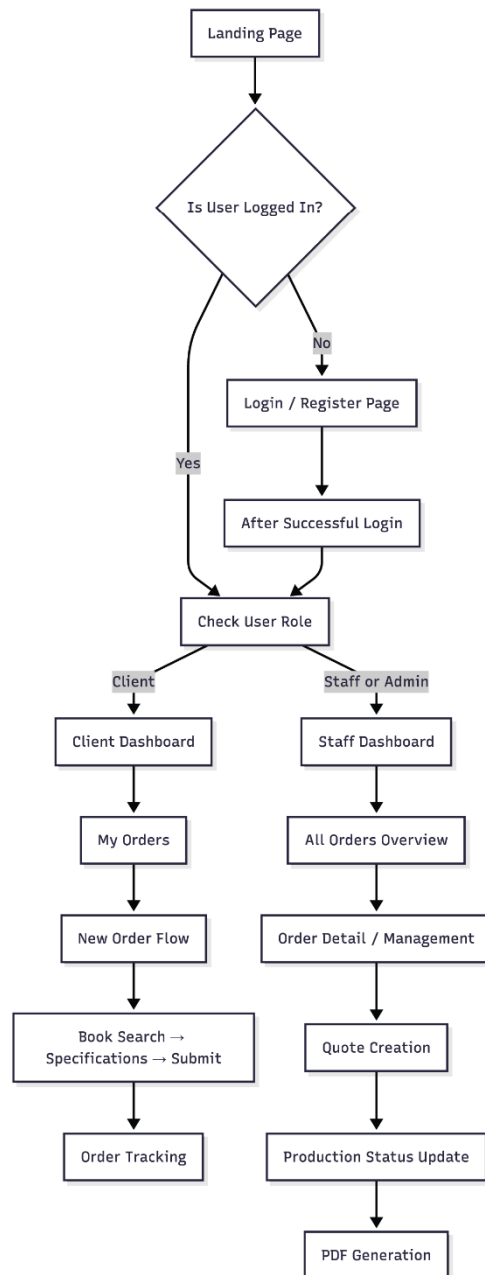
Authentication Logic:

1. User lands on the home page.
2. If not logged in → Redirect to Login/Register page.
3. If logged in → Redirect to the appropriate dashboard based on user role.

4. Route Structure

Route	Access	Description
/	Public	Landing Page
/login	Public	Login Page
/register	Public	Registration Page
/client/dashboard	Client	Client Dashboard (My Orders)
/client/new-order	Client	New Order Submission (multi-step)
/client/order/:id	Client	Order Details and Tracking
/staff/dashboard	Staff/Admin	Staff Overview Dashboard
/staff/orders	Staff/Admin	All Orders List + Filters
/staff/order/:id	Staff/Admin	Order Management View
/unauthorized	All	Access Denied Page
*	All	404 Not Found Page

5. Frontend Control Flow Diagram



6. Main User Flows

Client Flow:

- Landing Page → Login/Register → Client Dashboard → My Orders → New Order (Book Search → Specifications → Submit) → Order Tracking

Staff/Admin Flow:

- Login → Staff Dashboard → All Orders → Order Detail → Quote Creation → Production Updates → PDF Generation

7. Key Reusable UI Components

Layout Components:

- Navbar (role-based)
- Sidebar (Staff only)
- ProtectedRoute
- Footer

Feature Components:

- BookSearch (with Google Books API)
- MultiStepOrderForm
- OrderCard / QuoteCard
- StatusTracker (progress visualization)
- DataTable (sortable & filterable)
- PDFPreviewModal
- SignaturePad (stretch)

Utility Components:

- Toast Notifications
- Loading Spinner
- Confirmation Modal

8. Key Screen Descriptions

Landing Page

Hero section, clear value proposition, and prominent Login/Register buttons.

Client Dashboard

Overview of active orders, status summary cards, and quick “New Order” button.

New Order Page

Multi-step form:

1. Book Search (API + manual entry)
2. Binding Specifications
3. Review & Submit

Staff Dashboard

Summary statistic cards, recent orders, and quick navigation.

Order Management Page (Staff)

Full order details, editable quote section, production status updater, delivery records, and PDF generation button.

9. Stretch Goals (Frontend)

- PDF Quote/Invoice generation using jsPDF
- Digital signature capture
- Analytics dashboard with charts
- Real-time notifications

The information outlined in this document is a basic overview of the frontend specifications of the BindFlow application that will be created for Obras Digitales. Some of these might change due to time constraints or knowledge gaps that may be experienced along the completion process. However, the main functionalities and project objectives will remain unaltered.